Field-induced inhomogeneous index distribution of a nano-dispersed nematic liquid crystal metamaterial near the Freedericksz transition: Monte Carlo studies

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Monte Carlo studies of the field-induced complex refractive index changes in nano-dispersed nematic liquid crystals (NLCs) exhibiting negative–positive refractive indices have been performed for various cases of applied field strengths and anchoring conditions over a broad spectral regime. The resultant field-induced spatially inhomogeneous molecular order and the corresponding metamaterial properties are obtained for various wavelengths and applied field strengths below and above the Freedericksz transition. In general, the director axis reorientation and the resultant refractive index distribution are spatially inhomogeneous, even under a uniform applied field. The detailed computations have identified parameter sets for obtaining the negative index of refraction and maximal index modulations that can be more than an order of magnitude larger than those obtainable from pure NLC systems. © 2010 Optical Society of America

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1. INTRODUCTION

Metamaterials that exhibit complex refractive indices otherwise not possible with naturally occurring materials have attracted intense interest for fundamental pursuits as well as novel applications [1–12]. In particular, since the pioneering work of Veselago [1] more than 40 years ago, metamaterials possessing negative refractive indices have been proposed and demonstrated in microwave–optical regions. For some applications, tunability of the refractive index of the metamaterials is desirable. In the case of materials containing liquid crystals the tunability is enabled by the (electrical and/or optical) field-controlled birefringence of the NLC constituents [7–14]. In the two exemplary metamaterial systems under study, namely nanostructures with liquid crystal layers [6] and aligned NLCs containing randomly dispersed nanospheres [12], the response of the NLC host to the applied external field is taken to be uniform.

While such simplified analytical treatment enables clear illustration of the general results and trends, many features associated with nonlocality or inhomogeneous spatial dependence of field-induced reorientation of the liquid crystalline axis are lost. Yet such nonlocal inhomogeneous nematic responses are particularly important to obtain an accurate quantitative description of index modulation and other optical responses. Furthermore, in situations involving nonuniform applied fields, for example in photorefractivity-mediated optical wave mixing processes [14], these detailed accurate treatments often produce new insights and unexpected and potentially useful effects. The aim of this work is to calculate and analyze these nonlocal and spatial inhomogeneities of refractive index in the nanosphere dispersed liquid crystal (NDLC) metamaterial [12] system and to elucidate their dependence on the wavelength, anchoring force, and applied spatially modulated electric field. These quantitative calculations also enable us to identify parameter sets that will optimize the field-induced index modulation and other useful metamaterial properties. For this purpose, we have employed Monte Carlo simulation techniques, which have been demonstrated to be well suited for modeling such NLC responses, e.g., for chessboard NLC cell and twist alignment NLC displays [15,16].

2. THEORETICAL BACKGROUND

A. Coated Nanospheres Dispersed Nematic Liquid Crystals

In the formalism employed by [9,12], electromagnetic wave scattering from a single coated sphere is calculated using Mie theory [17], while the effective scattering from an ensemble of coated spheres dispersed in an NLC matrix is analyzed in effective medium theory using the Maxwell Garnet mixing rule [18,19]. The effective refractive index of an NDLC metamaterial is governed by the parameters of its constitutive components: coated nonmagnetic spheres and the host NLC system.

Figure 1 shows the makeup of a coated nanosphere dispersed in a host NLC. The core with radius r_1 is made from polaritonic material (such as $LiTaO_3$) with relative permittivity

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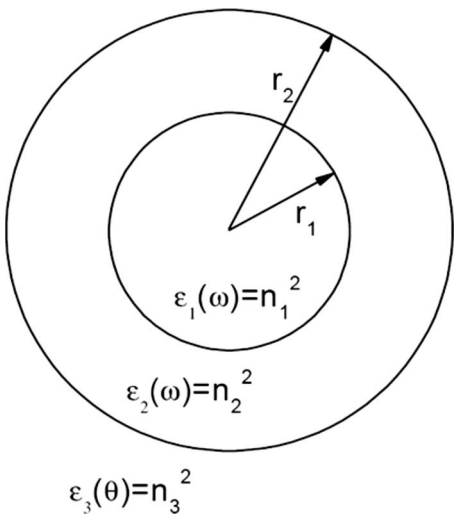

Fig. 1. Cross section of a coated nanosphere inside the liquid crystal host.

$$\varepsilon_1 = \varepsilon(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right). \tag{1}$$

Here $\varepsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 is the damping coefficient, and ω_T and ω_L are the transverse and longitudinal phonon frequencies, respectively. Bulk cavity resonance in the core is responsible for the negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths. The relative permittivity of the semiconductor shell with outer radius r_2 is described by a Drude model,

$$\varepsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p denotes the plasma frequency and γ_2 is the damping coefficient. The shell is responsible for the negative effective permittivity, with ω_p close to the frequency for which the negative index will occur. Its value is chosen to provide both negative effective permittivity and permeability in some interval of wavelengths.

Permittivity of the liquid crystal host for an extraordinary polarized wave depends on angle θ between directions of the incident light wave vector and the director axis of the NLC [12–14],

$$\varepsilon_3(\theta) = \frac{\varepsilon_\parallel \varepsilon_\perp}{\varepsilon_\parallel \cos^2\theta + \varepsilon_\perp \sin^2\theta}, \tag{3}$$

where $\varepsilon_\parallel$ and $\varepsilon_\perp$ are the respective permittivities for light polarized parallel and perpendicular, respectively, to the director axis. In a pure NLC host (without spheres), $k_3 = \sqrt{\varepsilon_3}k_0$.

The effective permittivity and permeability of the NDLC, calculated in effective medium theory in dipolar (electric and magnetic) approximations, are given by, respectively, [12]

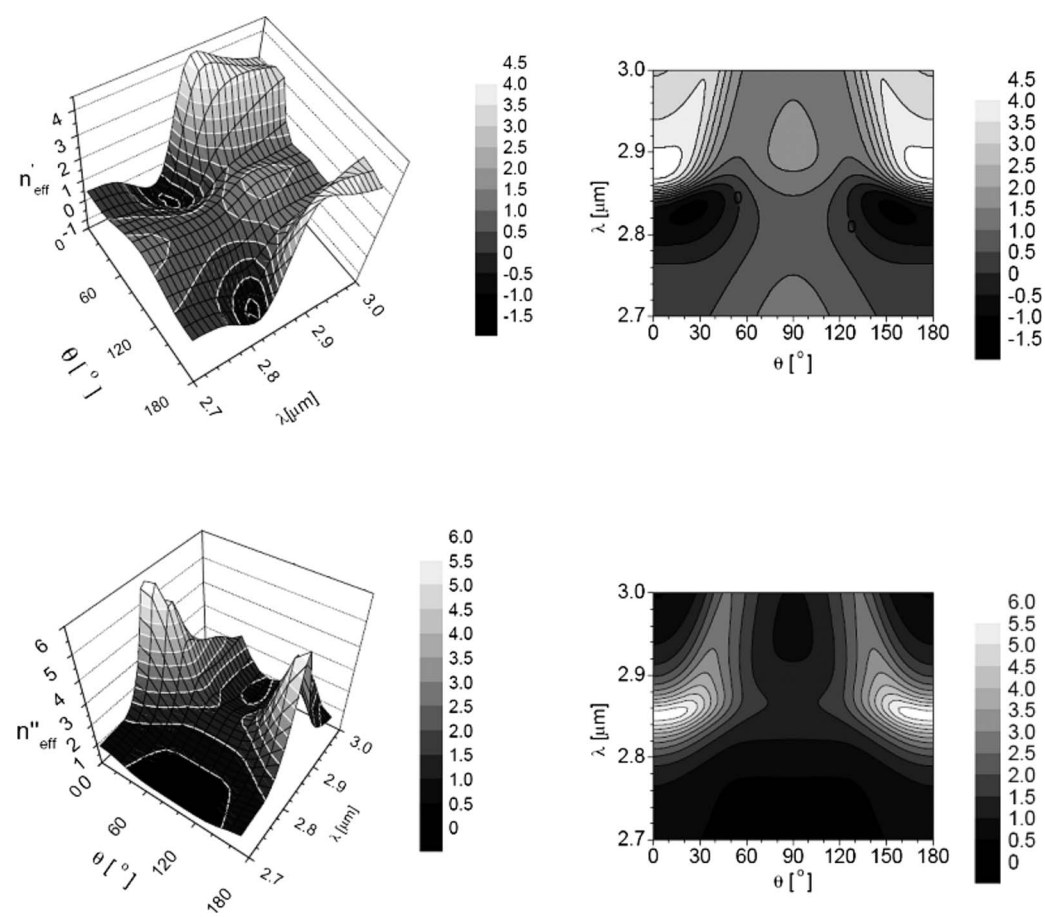

Fig. 2. Plots of real and imaginary parts of an effective refractive index: $n_{\mathrm{eff}}'(\lambda,\theta)$ (top) and $n_{\mathrm{eff}}''(\lambda,\theta)$ (bottom).

$$\varepsilon_{\text{eff}} = \varepsilon_3\left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1}\right), \quad \mu_{\text{eff}} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}. \tag{4}$$

Detailed expressions for the Mie scattering coefficients a_1 and b_1 can be found in [12,17]. Note that these expressions are valid only for small-number density $N = 3f/4\pi r_2^3$ of the nanospheres (f is the filling factor) and in long-wavelength limit $2\pi/\mathrm{k}_3 \gg \mathrm{r}_2$. The effective index of refraction is

$$n_{\text{eff}} = n_{\text{eff}}' + i n_{\text{eff}}'' = \pm\sqrt{\varepsilon_{\text{eff}}\mu_{\text{eff}}}, \tag{5}$$

where we have used the root corresponding to a positive imaginary part. Based on a homogeneous-response formalism, i.e., a uniform field-induced reorientation angle $\delta\theta$ throughout, Khoo *et al.* have shown [12] that for some values of the parameters of the model there is a range of frequencies ω and angles θ for which the real part of the effective refractive index $n_{\text{eff}}'(\omega,\theta)$ is negative.

B. Homogeneous Order: Constant Electric Field

For the purpose of comparing and contrasting the results from the (homogeneous) response formalism with the present treatment, we reexamine here some of the pertinent results from the former approach. We start by a detailed analysis of real (n'_{eff}) and imaginary (n''_{eff}) parts of the effective refractive index (5) as a function of wavelength λ of the incident light and of the angle θ, which the nematic axis makes with the $\hat{z}$ axis, assuming $\varepsilon_\parallel = 4$, $\varepsilon_\perp = 2$, i.e., the reorientation angles are uniformly distributed such that the director axis is parallel or perpendicular to the optical electric field. The exemplary plots in Fig. 2 illustrate some important features.

In the plot of n'_{eff} two separate areas of negative refractive index are present. The minimum (−1.5) occurs at $\lambda = 2820$ nm. A steep slope appears for longer wavelengths, with the maximum value of refractive index (+4.5) at $\lambda = 2890$ nm. The imaginary part n''_{eff} displays a strong narrow maximum for $\lambda = 2850$ nm, which corresponds to the steep slope of n'_{eff}. For wavelengths not shown in the plots the surfaces do not show any unusual interesting features. The strongest dependency on wavelengths occurs for angles close to 15° or 165°, while for angles in the vicinity of 90° the refractive index changes slowly.

In what follows we analyze four exemplary cases corresponding to the wavelengths $\lambda = 2820$, 2830, 2850, and 2900 nm. Figure 3 shows the respective dependence of n_{eff}' on angle θ. Despite the closeness in values of λ (2820, 2830, 2850 nm), the corresponding plots take on very different behaviors for those angles θ that correspond to the near parallelism of the local director and the $\hat{z}$ axis ($\theta < 25°$ or $\theta > 155°$). For linearly polarized light with $\lambda = 2820$ or 2830 nm incident as an extraordinary wave on the NLC host, one can see that $n_{\text{eff}}' < 0$. On the contrary, light with $\lambda = 2850$ nm propagates with a positive refractive index. For large values of angle $60° < \theta < 120°$ refractive indices are positive in all cases. The case $\lambda = 2900$ nm is different: n'_{eff} is always positive and has a surprisingly large value $n' \sim 4$ for θ close to 0° or 180°. These plots play the central role in the rest of the paper.

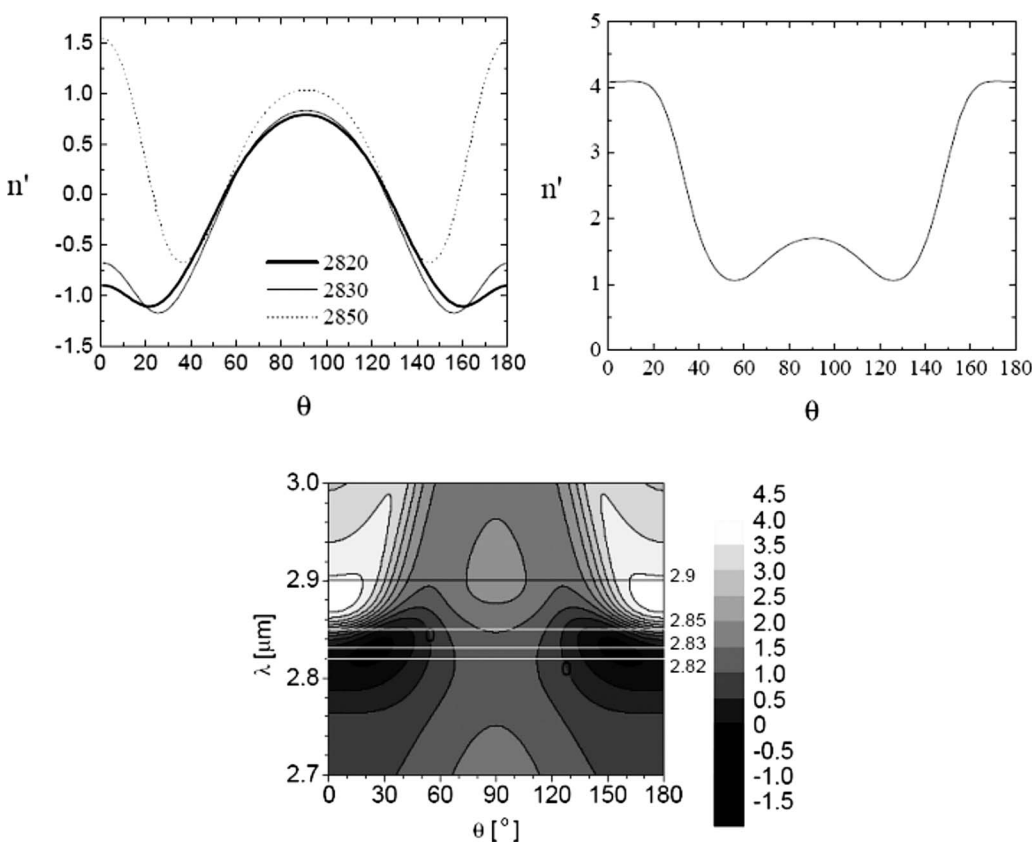

Fig. 3. Real part $n_{\text{eff}}'(\theta)$ of the local refractive index versus θ for wavelengths $\lambda = 2820$, 2830, 2850 nm (upper left), and 2900 nm (upper right). Contour plot of $n_{\text{eff}}'(\theta,\lambda)$ with lines corresponding to $\lambda = 2820$, 2830, 2850, and 2900 nm (bottom).

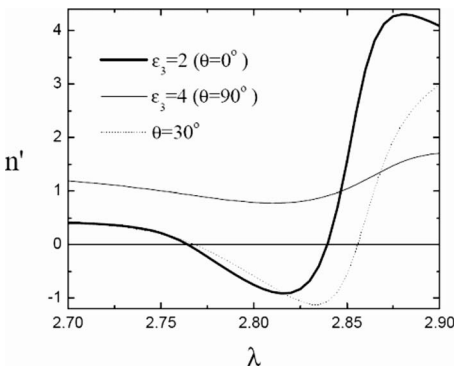

Fig. 4. n_{eff}' against λ in case of homogeneous order (θ=const).

Figure 4 summarizes the results in the case of the homogeneous order (i.e., θ=const throughout the bulk). Curves for angles $\theta=0,\pi/2$ are in agreement with those reported in [12]. Note that the widest wavelength range for which $n'_{\mathrm{eff}}<0$ corresponds approximately to $\theta=30°$. As we will see presently, when spatially inhomogeneous director axis reorientation is accounted for, considerable deviations from this picture will emerge.

3. MONTE CARLO SIMULATIONS

In our model the NLC molecules, represented by 1D rods, occupy sites of an $n_x\times n_y\times n_z$ rectangular lattice. Translational motions of the molecules are prohibited. A particular orientation of a long axis of an NLC molecule in an external coordinate system is described by a pair of angles φ,θ. Angle φ (twist) describes an azimuth, and angle θ (tilt) is a polar angle measured from the z axis, along which the electromagnetic wave propagates with polarization along the x axis. We apply rubbing along the x axis. The molecules lie (and fluctuate) in the x–z plane. Along the x and y directions the periodic boundary conditions are applied, while the molecules in the lowest (1) and the uppermost (n_z) planes are subjected to properly chosen anchoring forces due to their interaction with a substrate.

Interactions are described by the Lebwohl–Lasher effective Hamiltonian [20] with the Rapini–Papoular term [21] in units k_BT (k_B, Boltzmann constant; T, absolute temperature):

$$H=-\xi\sum_{\langle\vec{r},\vec{r}'\rangle}P_2(\cos\beta(\vec{r},\vec{r}'))-\sum_{\vec{r}}E^2(\vec{r})P_2(\cos\beta(\vec{r}))+\sum_{\vec{r}_w}\alpha(\vec{r}_w)\sin^2\gamma(\vec{r}_w). \quad (6)$$

Here P_2 denotes the second-order Legendre polynomial. Parameter ξ (in simulation ξ=25) is the coupling constant for NLC–NLC orientational interaction, $E(\vec{r})$ is proportional to external electric field, and $\alpha(\vec{r}_w)$ denotes the amplitude of anchoring forces acting on molecules located at the walls. Parameters ξ,α,E^2 are given in units of k_BT. $\beta(\vec{r},\vec{r}')$ denotes the relative angle between two molecules located at points $\vec{r},\vec{r}'$. $\beta(\vec{r})$ is the angle that a molecule located at point $\vec{r}$ makes with the electric field direction. $\gamma(\vec{r}_w)$ is the angle between the long axis of the molecule at the wall and a fixed rubbing direction (Fig. 5). Each molecule placed at the site is surrounded by the six nearest neighbors, and interactions with those neighbors are taken into account.

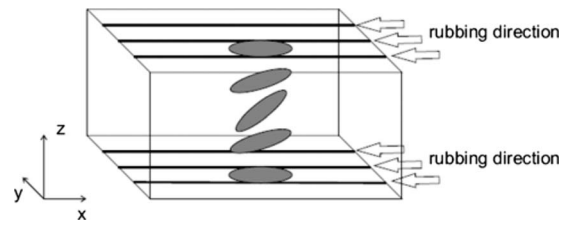

Fig. 5. Geometry of the simulated system. Electric field is along the z axis.

Consider the case of a spatially modulated applied electric field consisting of a constant field E_0 and a sinusoidal modulated part:

$$E(x)=E_0+A\sin(qx). \quad (7)$$

Spatial modulation of an external applied electric field and spatial modulation of the strength of the anchoring forces, the later deterministic or random in nature, have a strong impact on the resulting inhomogeneous order in a planar NLC cell [15,16]. They determine the spatial distribution of refractive index that, for some values of parameters, can assume negative values. Since a negative refraction index requires a fine tuning of the parameters of an NDLC [12], the specific details of the local-order inhomogeneity can play a decisive role for its coming into existence and, in particular, for the character of its spatial organization.

This modulation may be a result of sinusoidally modulated light intensity in a two-wave mixing experiment in a photorefractive NLC where the sinusoidal light intensity pattern produces a respective modulation of the electric field in the bulk or on surfaces [14,22,23]. The equilibrium configurations of NLC molecules on a lattice were sampled using Metropolis Monte Carlo simulations [24,25]. The system was equilibrated during 3×10^4 Monte Carlo steps (MCS) and then simulated during 2×10^5 MCS. From 2×10^5 equilibrium configurations generated during simulation, 2×10^2 MCS were used (each 10^3 MCS) to calculate the average values of chosen physical properties. In this paper we put $A=0.08$ (in unit of E_{f}, i.e., the magnitude of A is ~ a few percent of the Freedericksz field strength), and E_0 is near the Freedericksz threshold (for the case of a hard-boundary condition). In the simulations we have used the parameters similar to [12]: $r_1=0.13$ μm, $r_2=0.143$ μm, filling factor (nanosphere concentration by volume) $f=0.12$, $\varepsilon(\infty)=17$, $\omega_T/2\pi=240$ THz, $\omega_L/2\pi=570$ THz, $\gamma_1/2\pi=2.5$ THz, $\omega_p/2\pi=134$ THz, and $\gamma_2=\omega_p/60$.

A. Inhomogeneous Order: Uniform Applied Electric Field

In this section we discuss inhomogeneous order in the case when an external electric field is constant: $\vec{E}(\vec{r})=E\hat{z}$ and E exceeds the Freedericksz threshold E_F [14]; we also assume that anchoring forces are spatially homogeneous: $\alpha(\vec{r}_W)=\alpha$=const. In the calculations we have neglected, for simplicity, the azimuthal angular variation (φ) and have

taken into account only polar angle θ. Even in this simple case an inhomogeneous metamaterial resulted.

Figure 6 depicts two-dimensional maps of $n'_{\mathrm{eff}}(x,y,z)$ averaged over the y axis, calculated in the strong anchoring regime, with $\alpha=50$ and for electric field $E=1.2>E_F$. Regions marked with a striped pattern have negative refractive indices. The distributions do not depend on the x coordinate because of the absence of electric field modulation along x. After averaging over x and y directions we get profiles $n'_{\mathrm{eff}}(z)$ presented in Fig. 6 [26]. The NDLC displays the layered structure with layers of normal material ($n'_{\mathrm{eff}}(z)>0$) and of metamaterial (with $n'_{\mathrm{eff}}(z)<0$. The spatial organization of the layers strongly depends on the incident light wavelength. For $\lambda=2.820$ μm, we find three layers with the central one displaying metamaterial properties. In the case of $\lambda=2.850$ μm we observe five layers; two of them correspond to a metamaterial.

Figure 7 shows the dependence of profiles $n'_{\mathrm{eff}}(z)$ on wavelength λ. Below the Freedericksz transition threshold, i.e., $E_0<E_F$, the (real part of) effective refractive index $n'_{\mathrm{eff}}(z)>0$. For $E_0>E_F$ the single-domain metamaterial (with $n'_{\mathrm{eff}}(z)<0$) pattern gives way smoothly to a double-domain metamaterial pattern and then gradually gets transformed into a normal material (with $n'_{\mathrm{eff}}(z)>0$) as the wavelength increases.

B. Field-Induced Inhomogeneous Nematic Order-Modulated Electric Field

In this section we discuss the inhomogeneous spatial structure of the refractive index in NDLC resulting from spatial modulation (along the x axis) of electric field [cf., (7)]. A typical x-z configuration of NLC molecules in strong anchoring regime is shown in Fig. 8. Molecules close to the plates remain flat, while those in the bulk are rotated, depending on local strength of field $E(x)$. In what follows we focus our attention on two cases: $\lambda=2900$ nm and $\lambda=2830$ nm.

Case A: Positive-Index Material

We analyze the case $\lambda=2900$ nm. Refractive index n'_{eff} is always positive and in the range from 1 to 4 (see Figs. 9 and 10). Let us discuss the strong anchoring regime first. For a weak electric field, spatial distribution of a refractive index is almost homogeneous. It has low value $n_{\mathrm{eff}}\approx 1$. The strongest changes occur in the middle of the cell, while at the boundaries the molecules are strongly aligned and therefore index changes are small. For a strong field ($E_0=1.2$) the amplitude of changes of a refractive index in the x direction is small. Molecules far from walls are aligned along the field because the local field

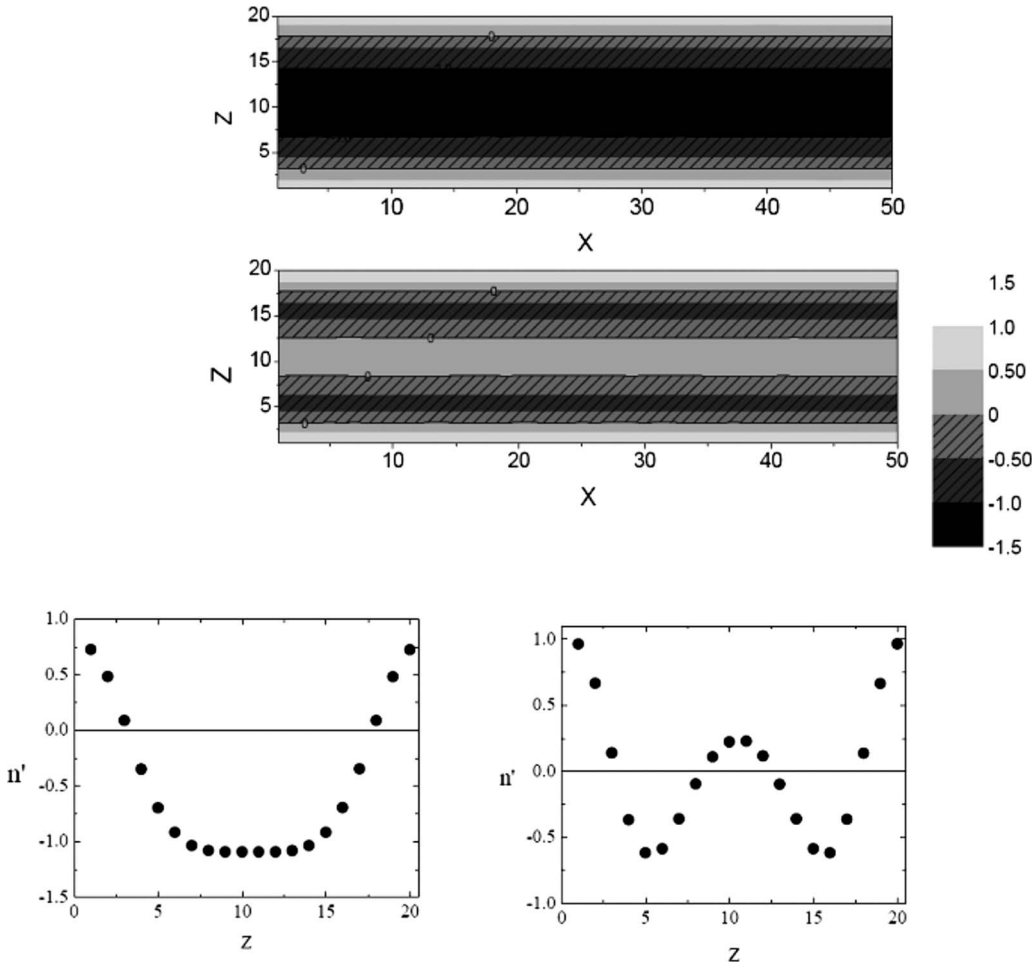

Fig. 6. 2D maps of $n'_{\mathrm{eff}}(x,z)$. Regions marked with striped pattern have negative values of refractive index: $\lambda=2.820$ μm (top), $\lambda=2.850$ μm (bottom). Profiles $n'_{\mathrm{eff}}(z)$: $\lambda=2.820$ μm (lower left), $\lambda=2.850$ μm (lower right). $E_0=1.2$, $\alpha=50$ (after [26]).

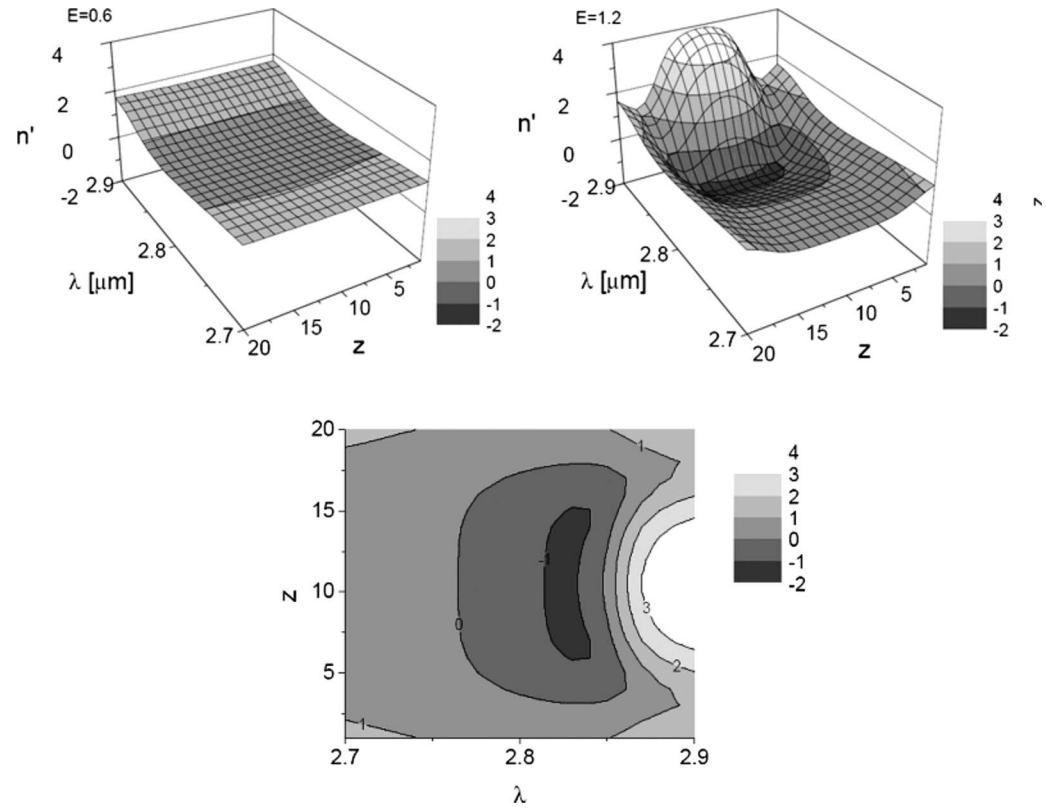

Fig. 7. Profiles $n'_{\mathrm{eff}}(z)$ in function of wavelength λ in strong anchoring regime $\alpha=50$, $E_0=0.6$ (upper left), $E_0=1.2$ (upper right and bottom).

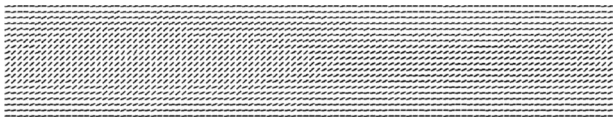

Fig. 8. Cross section in the x–z plane of a typical spatially modulated 3D configuration of NLC particles ($E_0=1.2$, $\alpha=50$).

$E(x)$ considerably exceeds E_F. We note that between $E_0=0.8$ and $E_0=1.0$ a kind of "inversion" feature takes place; under weak fields ($E_0<0.8$), the area $x<50$ where the field is the maximum exhibits a lower refractive in-

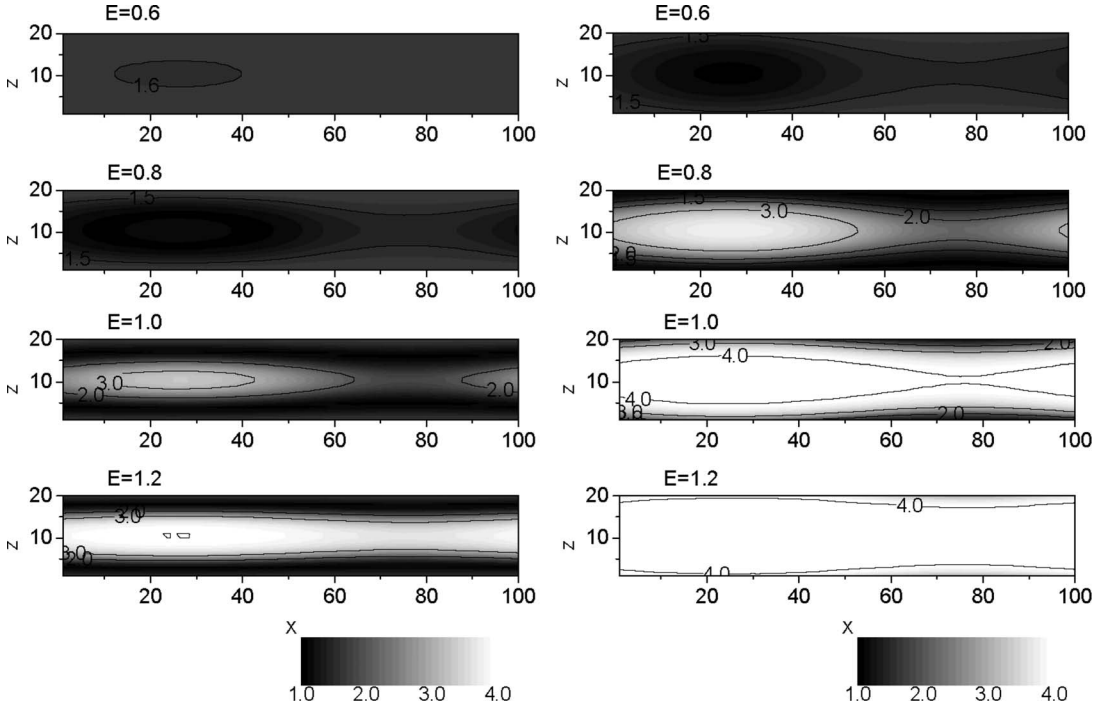

Fig. 9. 2D maps of $n'_{\mathrm{eff}}(x,z)$ for $\lambda=2900$ nm, in strong ($\alpha=50$, left) and weak ($\alpha=10$, right) anchoring regimes for a few values of E_0 shown in the figures.

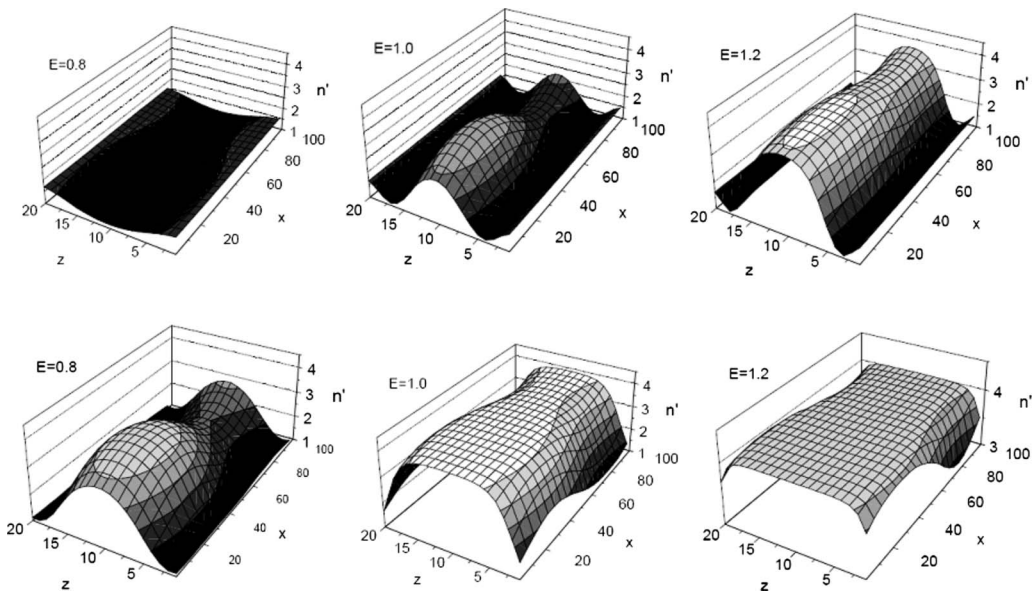

Fig. 10. Plots of $n'_{\mathrm{eff}}(x,z)$ for $\lambda=2900$ nm, in strong ($\alpha=50$, top) and weak ($\alpha=10$, bottom) anchoring regimes for a few values of E_0 shown in the figures.

dex, whereas in the case of stronger applied fields ($E_0>1$), these islands exhibit high index values.

Consider now the weak anchoring regime. In this case the system shows more "flexibility" for index modulation. Even under a low applied electric field ($E_0=0.8$), the index distribution is not uniform. The index of refraction changes not only in the middle of the cell but also near the walls, as one may expect from the weak anchoring condition. The largest index amplitude modulation $\Delta n'_{\mathrm{eff}}$, i.e., the biggest difference between index values in different regions, occurs when the field is relatively weak ($E_0=0.8$). For a strong field ($E_0=1.2$), distribution of the refractive index is almost homogeneous; nevertheless, near the walls n'_{eff} also undergoes large changes.

Let us discuss next the averaged index $<n'>(x)$, which governs the phase shift of light transmitted through cell of thickness L along the z direction [26]:

$$\langle n'\rangle(x)=\frac{1}{L}\int_0^L n'_{\mathrm{eff}}(x,z)\mathrm{d}z. \tag{8}$$

Some exemplary results obtained from the computations, assuming a strong anchoring condition, are shown in Fig. 11. For $E_0=0.6$-0.8, the averaged index becomes smaller, the modulation $\Delta<n'>$ is opposite to the modulation of the electric field, and the amplitude of modulation increases. For $E_0=0.8$-0.9, we observe the inversion of phase. For fields higher than 0.9, the average value of the refractive index starts to increase. Modulation $\Delta<n'>$ is first an increasing and then a slightly decreasing function. The highest value of modulation occurs at $E_0=1.0$.

In the weak anchoring case, even for weak field $E_0=0.6$, the profile is not flat. Inversion of phase is observed between $E_0=0.6$ and $E_0=0.7$. The largest modulation occurs at $E_0=0.8$. For high fields the amplitude of modulation decreases dramatically. Figure 12 shows the plot of $\langle n'\rangle(x)$ for $\lambda=2900$ nm and $E_0=0.85$ in a weak anchoring case ($\alpha=10$) calculated for an NDLC (upper curve) and a pure NLC cell (lower curve). The index amplitudes read $\Delta\langle n'\rangle_{\mathrm{NDLC}}=1.1$ and $\Delta\langle n'\rangle_{\mathrm{NLC}}=0.1$. Since diffraction efficiency is proportional to index amplitude squared, $\eta\propto(\Delta n)^2$, diffraction efficiency for NDLC grating can be

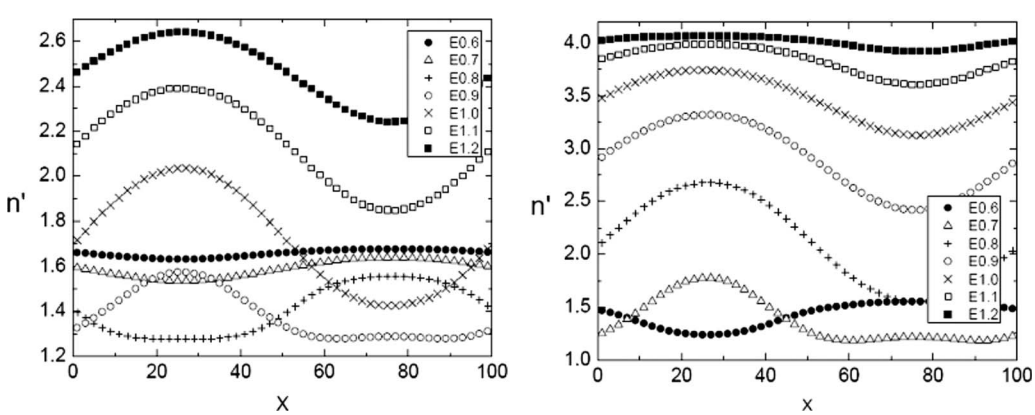

Fig. 11. Averaged profiles $\langle n'\rangle(x)$ for $\lambda=2900$ nm in strong ($\alpha=50$, left) and weak ($\alpha=10$, right) anchoring regimes for a few values of E_0 shown in the legends.

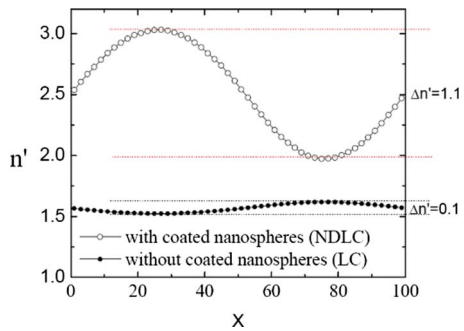

Fig. 12. (Color online) Modulation of refractive index for $\lambda = 2900$ nm for NDLQ (upper) and NLC cell (lower) for weak anchoring case ($\alpha = 10$), $E_0 = 0.85$.

more than 10^2 times larger than those obtainable from a pure NLC system.

Case B: Metamaterial (Negative Index)

At a shorter wavelength—2830 nm—the system displays different features (Fig. 13 and 14). Both positive and negative values of the refractive index are found. It ranges from −0.6 to 0.8 for strong anchoring and from −1 to 0.8 for weak anchoring.

First we discuss a strong anchoring regime. Under a weak field, the index distribution is almost homogeneous and the index is positive. For $E_0 = 0.9$ the refractive index in some regions becomes negative. Near the walls the index of refraction is always positive due to the planar alignment of molecules caused by strong anchoring. In regions where the electric field is strong the refractive index

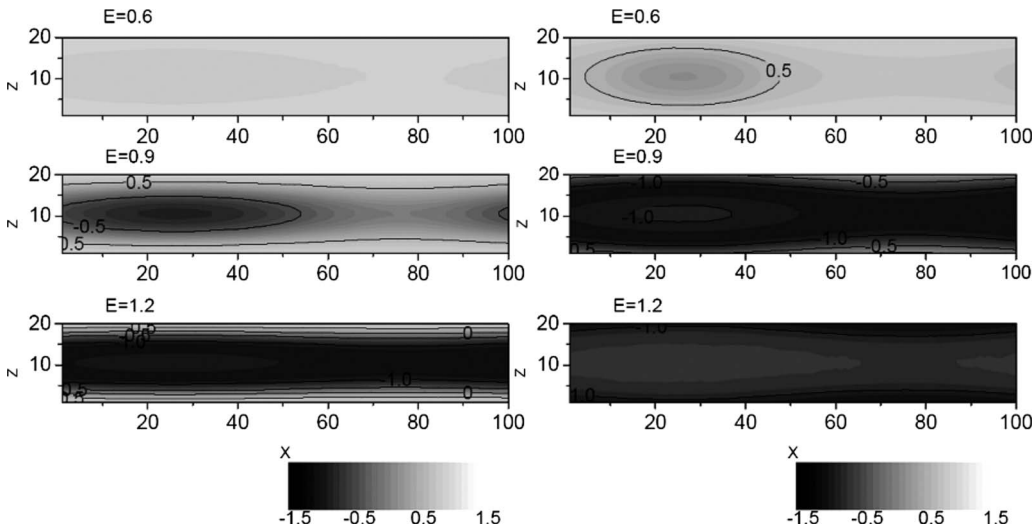

Fig. 13. 2D maps of $n'_{\mathrm{eff}}(x,z)$ for $\lambda = 2830$ nm in strong ($\alpha = 50$, left) and weak ($\alpha = 10$, right) anchoring regimes for a few values of E_0 shown in the figures.

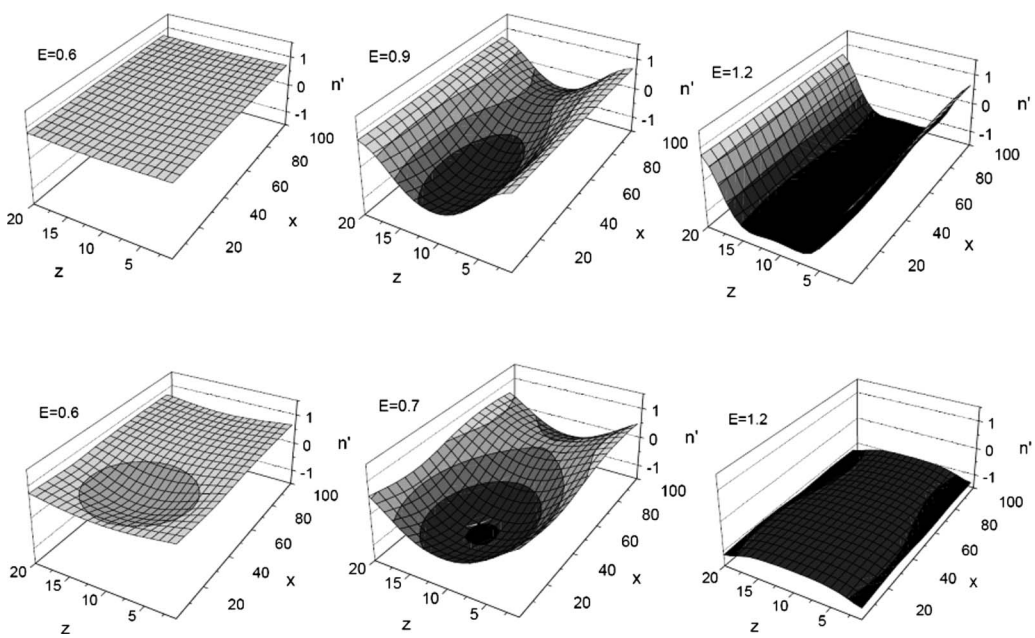

Fig. 14. Plots of $n'_{\mathrm{eff}}(x,z)$ for $\lambda = 2830$ nm in strong ($\alpha = 50$, top) and weak ($\alpha = 10$, bottom) anchoring regimes for a few values of E_0 shown in the figures.

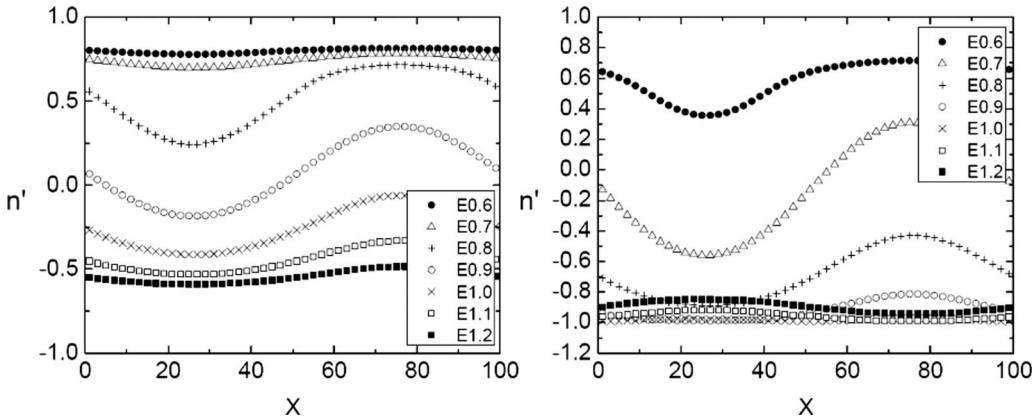

Fig. 15. Averaged profiles $\langle n' \rangle(x)$ for $\lambda = 2830$ nm in strong ($\alpha = 50$, left) and weak ($\alpha = 10$, right) anchoring regimes for a few values of E_0 shown in the legends.

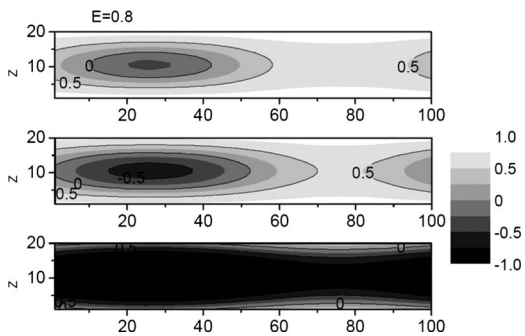

Fig. 16. 2D maps of $n'_{\mathrm{eff}}(x,z)$ for $\lambda = 2830$ nm and $E_0 = 0.8$ in three anchoring regimes: strong ($\alpha = 50$, top), medium ($\alpha = 30$, middle), and weak ($\alpha = 10$, bottom).

has lowest negative values. For a strong field ($E_0 = 1.2$), the index of refraction in the center of the cell is negative and does not depend on the x coordinate.

In a weak anchoring regime for a low electric field ($E_0 = 0.6$), the refractive index is positive in the whole cell and displays modulation along the x axis. A maximum amplitude of modulation of the refractive index occurs for field $E_0 = 0.7$. For a strong field ($E_0 = 1.2$), distribution of the index is almost flat and the index is negative ($n_{\mathrm{eff}} \approx -1$). Contrary to strong anchoring, the index near the walls has lower negative values than in the center.

Figure 15 presents the profiles of $<n'>(x)$ for chosen values of electric field E_0 in the case of strong and weak anchoring. We find that the largest $\Delta <n'>$ occurs at $E_0 = 0.9$ for strong anchoring and $E_0 = 0.7$ for weak anchoring. Finally, we discuss briefly the role of anchoring forces. Figure 16 depicts some of the salient dependence of the refractive index (at $\lambda = 2830$ nm) distribution on the anchoring force. For illustration, a moderate electric field $E_0 = 0.8$ has been chosen. We find that with a decreasing anchoring force the overall value of the refractive index becomes more negative, and the islands of negative index become broader.

4. CONCLUSION

We have presented a quantitative analysis of the effect of an applied field on a metamaterial comprised of nanospheres dispersed in an aligned NLC. We have shown that the spatial structure of a model NDLO metamaterial under experimental conditions, i.e., in presence of anchoring forces and an external electric field exceeding the Freedericksz threshold, leads necessarily to its inhomogeneous spatial structure. This structure, determined by anchoring forces and the field, is very sensitive to the wavelength and can result in large values of index amplitude variation in the case of a spatially modulated external field. In the case of a spatially modulated applied electric field, we have identified a parameter set that will create refractive index modulation whose magnitude can be orders of magnitude larger than a pure NLC system. The formalism developed here for planar NLC samples could also be applied to NLC cells with spatially modulated anchoring forces or more complex periodic nanostructures that exhibit metamaterial properties.

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